RP235xB

File: PGA2350B.kicad_sch

Ethernet

File: Ethernet.kicad_sch

Stepper Output

File: StepperOutput.kicad_sch

Misc IO

File: MiscIO.kicad_sch

Spindle

File: Spindle.kicad_sch

Limit and Control inputs

File: LimitandControlInputs.kicad_sch

DigitalIO and Relays

File: DigitalIO and Relays.kicad_sch

Power

File: power.kicad_sch

Servo_error

File: servo_error.kicad_sch

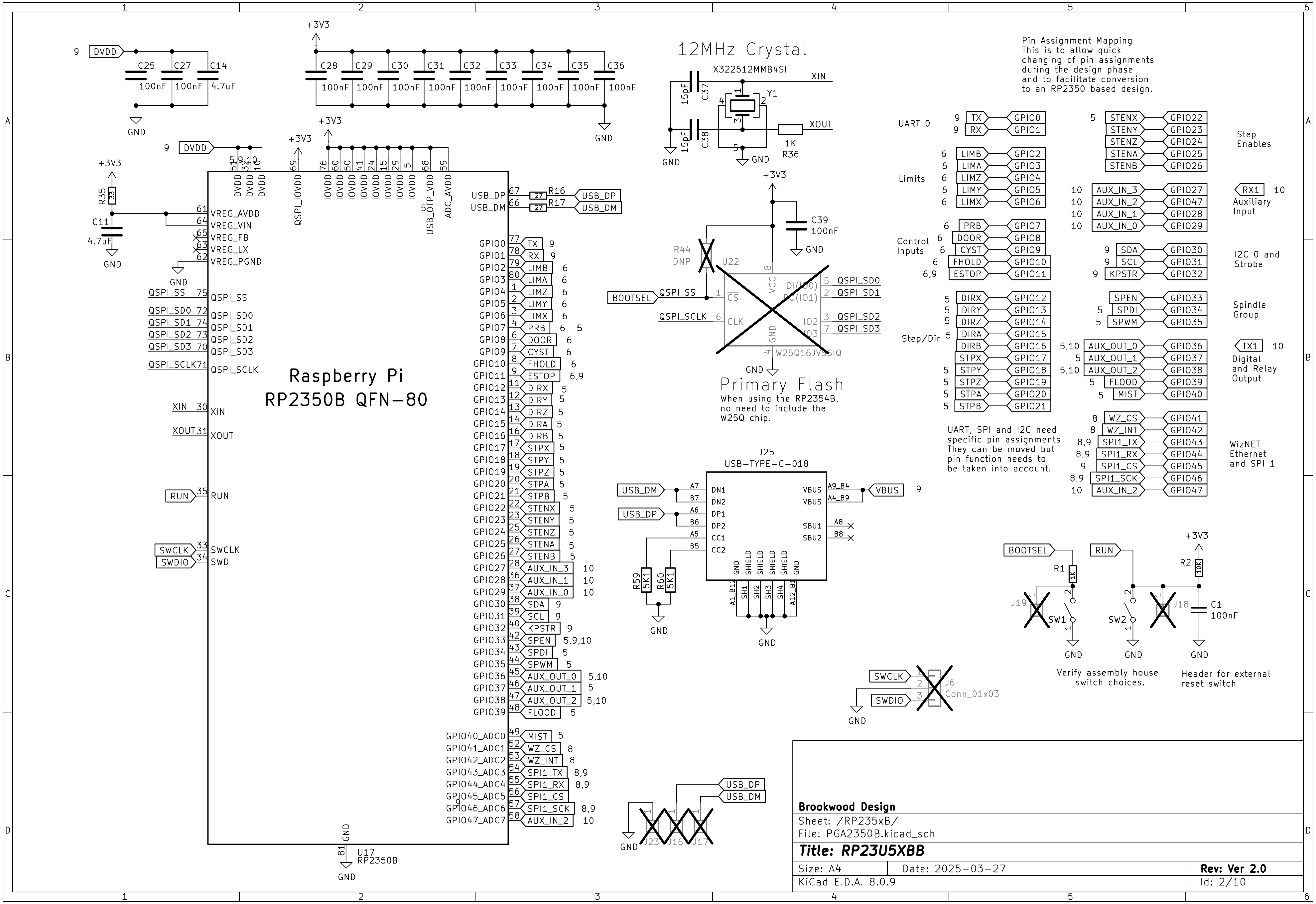
Brookwood Design

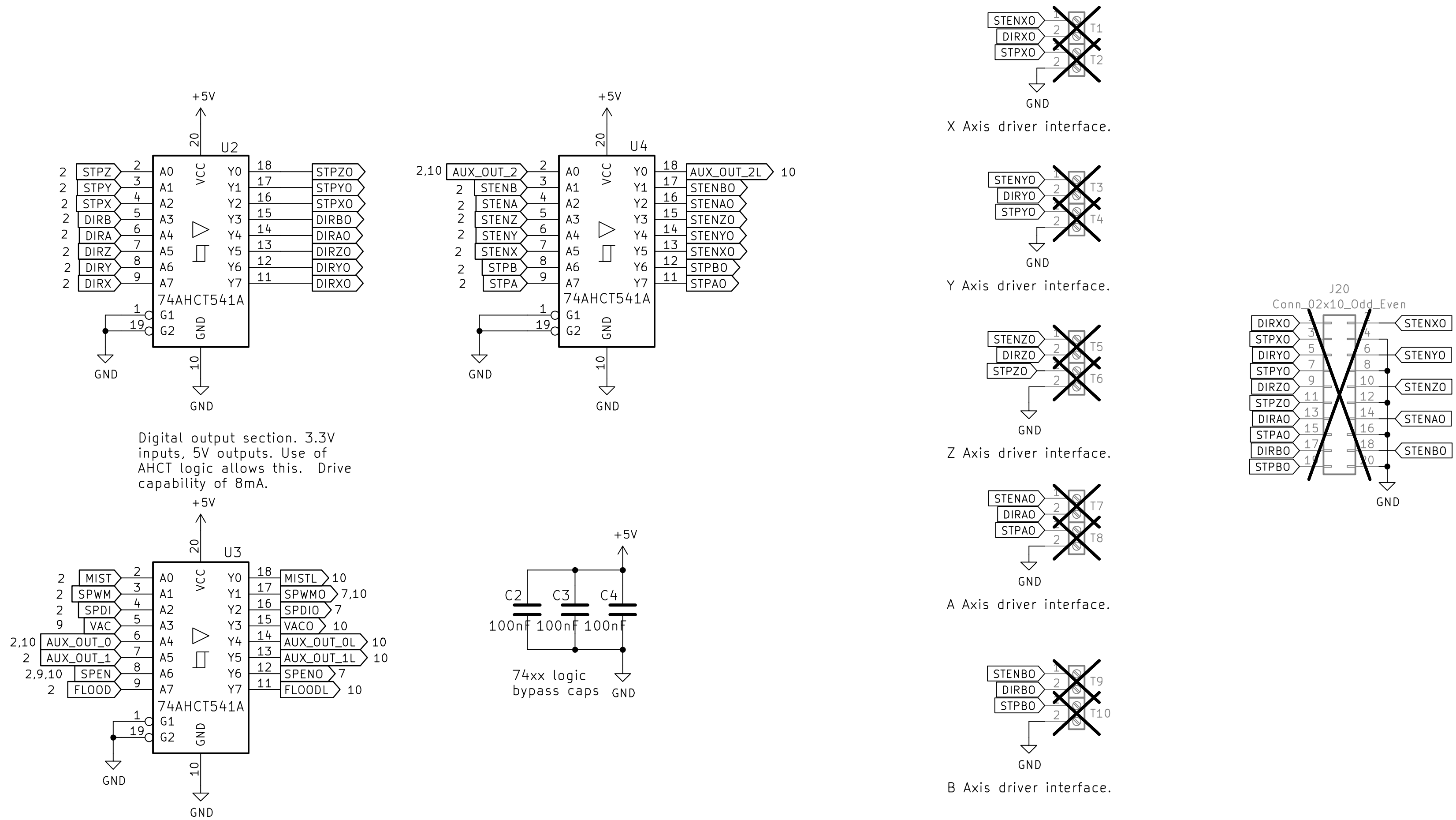
Sheet: /

File: v2.02.kicad_sch

Title: RP23U5XBB

Size: A4	Date: 2026-05-17	**Rev: Ver 2.01**
KiCad E.D.A. 8.0.9		Id: 1/10


12MHz Crystal
X322512MMB4SI
XIN
XOUT
Pin Assignment Mapping
This is to allow quick
changing of pin assignments
during the design phase
and to facilitate conversion
to an RP2350 based design.
UART 0
Limits
Control
Inputs
Step/Dir
Step
Enables
Auxiliary
Input
I2C 0 and
Strobe
Spindle
Group
Digital
and Relay
Output
WizNET
Ethernet
and SPI 1
UART, SPI and I2C need
specific pin assignments
They can be moved but
pin function needs to
be taken into account.
Raspberry Pi
RP2350B QFN-80
U17
RP2350B
Primary Flash
When using the RP2354B,
no need to include the
W25Q chip.
U22
W25Q16JVSSIQ
R44
DNP
J25
USB-TYPE-C-018
Verify assembly house
switch choices.
Header for external
reset switch
J6
Conn_01x03
Brookwood Design
Sheet: /RP235xB/
File: PGA2350B.kicad_sch
Title: RP23U5XBB
Size: A4
Date: 2025-03-27
Rev: Ver 2.0
KiCad E.D.A. 8.0.9
Id: 2/10



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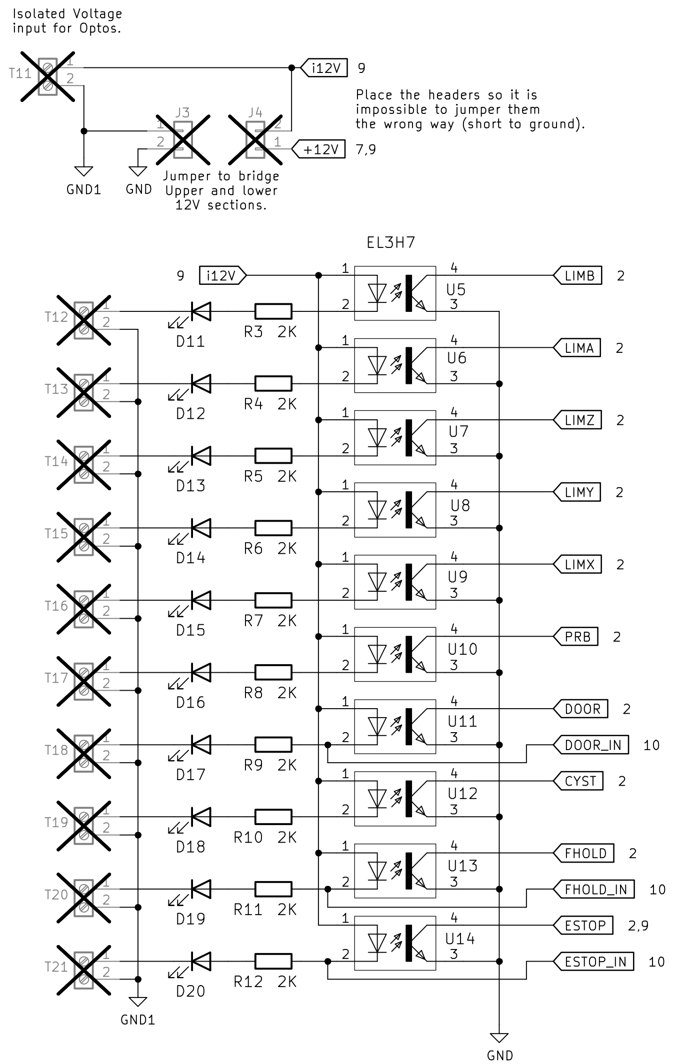
Sheet: /Stepper Output/

File: StepperOutput.kicad_sch

Title: RP23U5XBB

Size: A4	Date: 2025-03-27	**Rev: Ver 2.0**
KiCad E.D.A. 8.0.9		Id: 5/10



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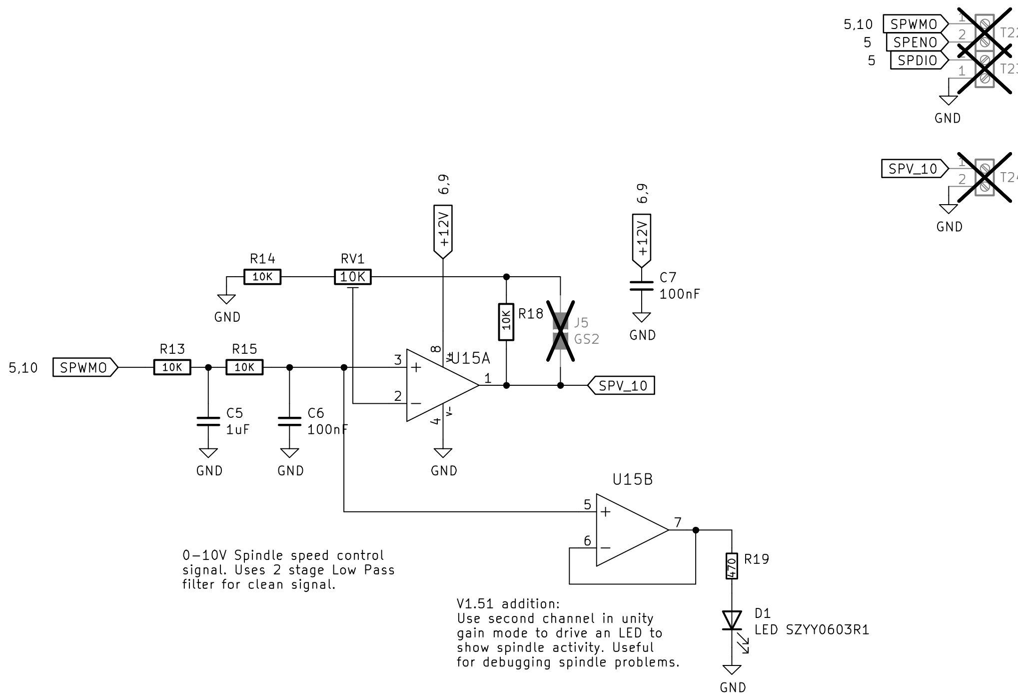
Sheet: /Limit and Control inputs/
File: LimitandControlInputs.kicad_sch

Title: RP23U5XBB

Size: A4 | Date: 2025-03-27 | **Rev: Ver 2.0**

KiCad E.D.A. 8.0.9 | Id: 6/10



Brookwood Design

Sheet: /Spindle/

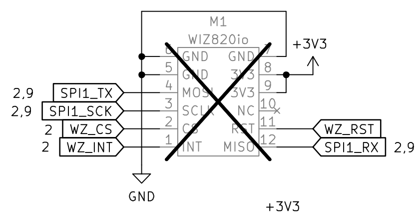
File: Spindle.kicad_sch

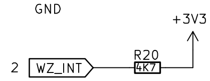
Title: RP23U5XBB

Size: A4	Date: 2025-03-27	**Rev: Ver 2.0**
KiCad E.D.A. 8.0.9		Id: 7/10

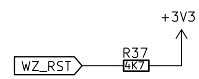
Wiz820io used here is technically the Wiz850io. Marketed as being compatible with the Wiz820io, it uses the W5500 chip. The Wiz820io uses the W5200 chip and is not supported. The ones called Wiz850-lite (Chinese knockoffs of the Korean original) are also supported.

Yes, it is confusing – just make sure the module has a W5500 on it and 2 6x1 pin headers on either side.





The WZ850io datasheet shows a pullup resistor on it but there have been reports of instability when WZ_INT does not have the additional pullup resistor. So, included out of caution – the additional pullup does not seem to hurt.



RST is held high. Need to make sure that there will not be a case where RST (active low) is needed. Evaluate if a higher value (10K) is better.

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Sheet: /Ethernet/

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Size: A4	Date: 2025-03-27	**Rev: Ver 2.0**
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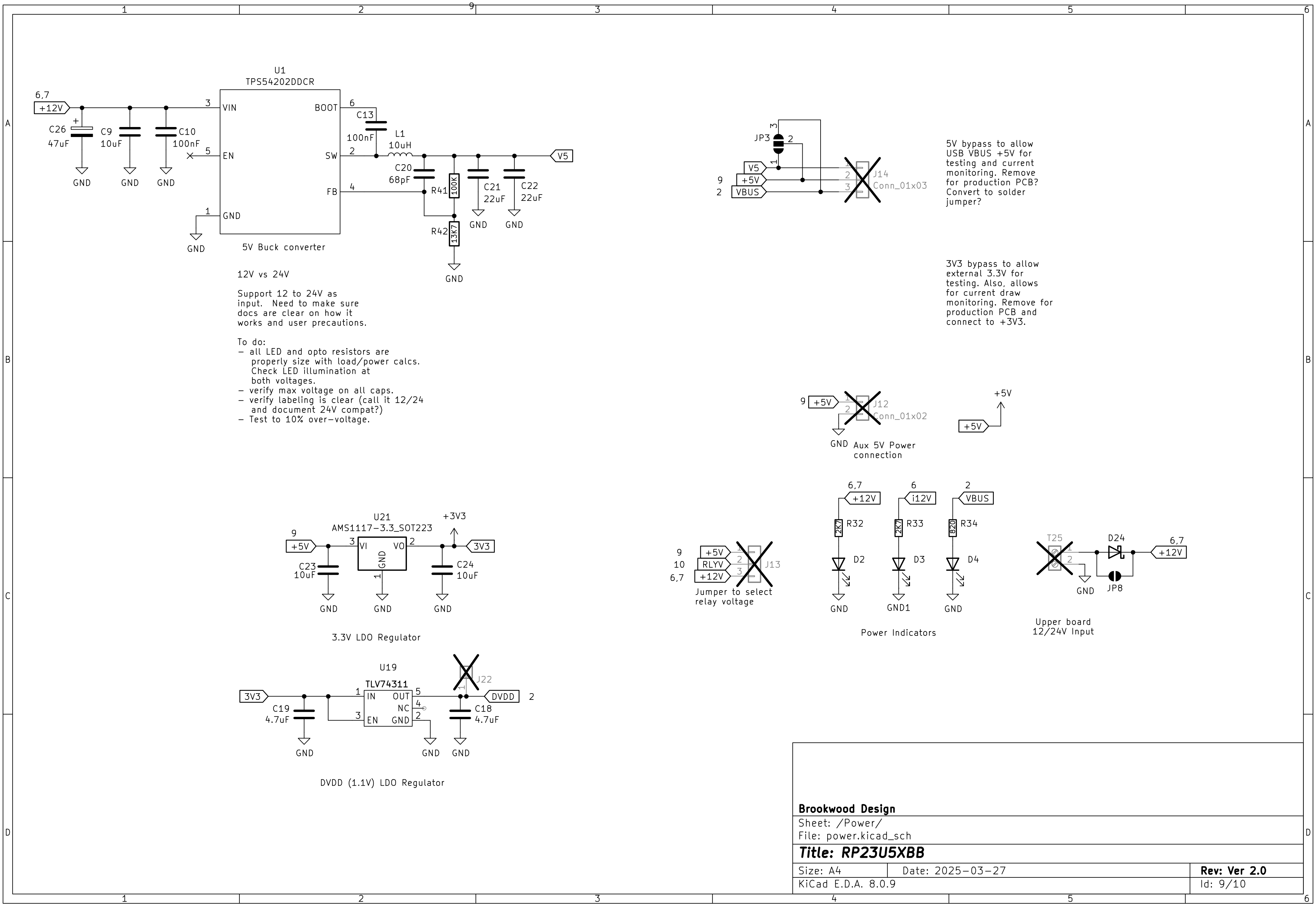

5V bypass to allow USB VBUS +5V for testing and current monitoring. Remove for production PCB? Convert to solder jumper?

3V3 bypass to allow external 3.3V for testing. Also, allows for current draw monitoring. Remove for production PCB and connect to +3V3.

12V vs 24V

Support 12 to 24V as input. Need to make sure docs are clear on how it works and user precautions.

To do:
- all LED and opto resistors are properly size with load/power calcs. Check LED illumination at both voltages.
- verify max voltage on all caps.
- verify labeling is clear (call it 12/24 and document 24V compat?)
- Test to 10% over-voltage.

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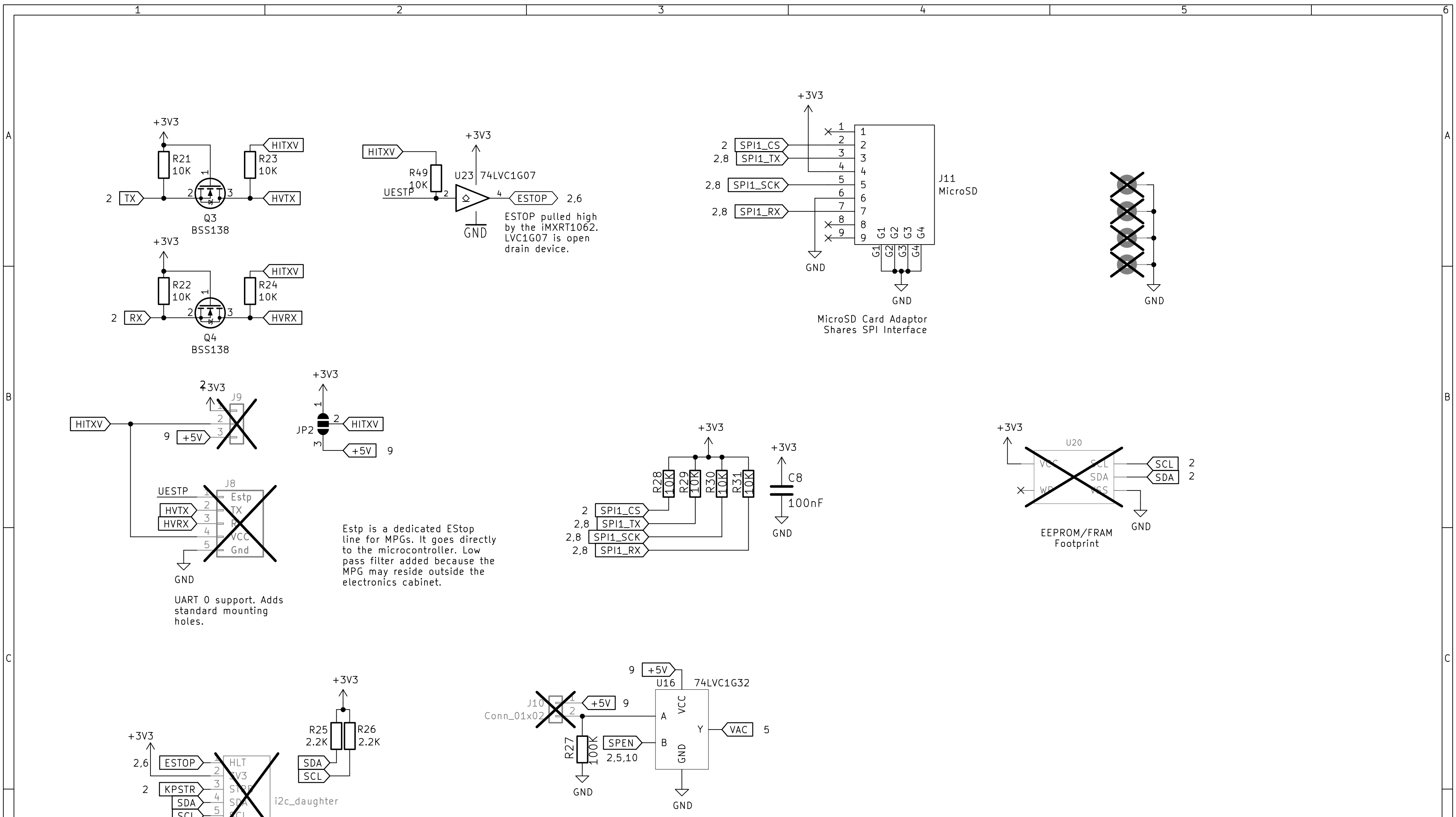
Sheet: /Power/

File: power.kicad_sch

Title: RP23U5XBB

Size: A4 | Date: 2025-03-27 | **Rev: Ver 2.0**

KiCad E.D.A. 8.0.9 | Id: 9/10



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Sheet: /Misc IO/

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Title: RP23U5XBB

Size: A4	Date: 2025-03-27	**Rev: Ver 2.0**
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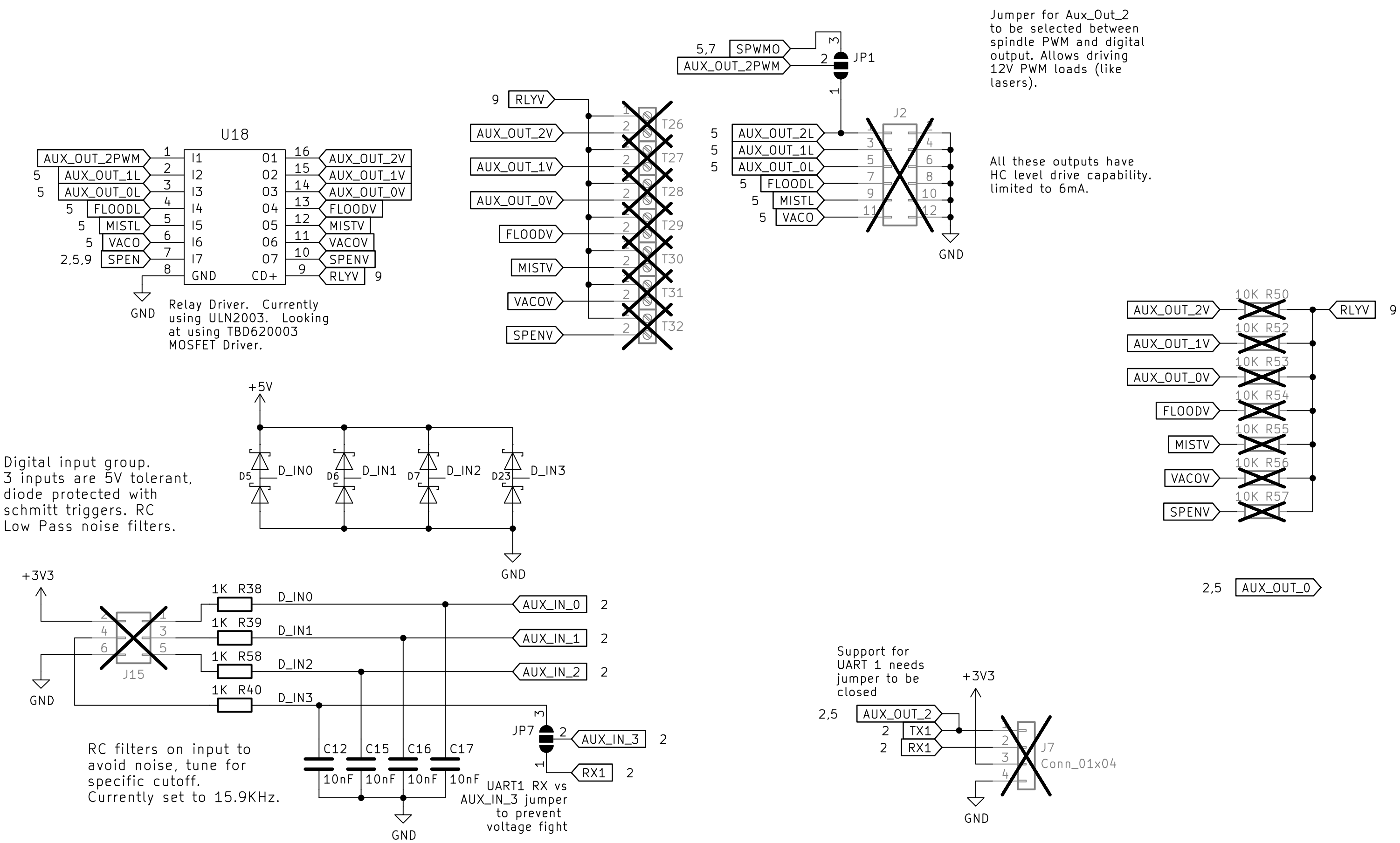

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Sheet: /DigitalIO and Relays/

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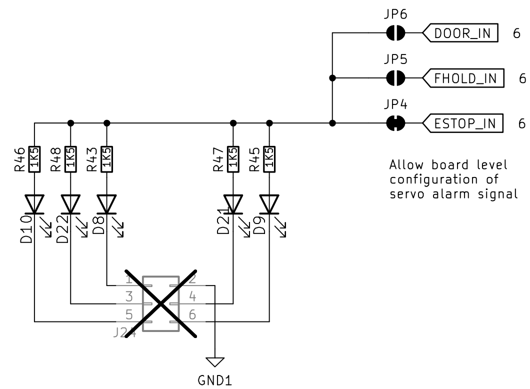
KiCad E.D.A. 8.0.9		Id: 10/10



Brookwood Design

Sheet: /Servo_error/

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